

ECE 113 Project Report

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Part 1: From Basic Time-Frequency Analysis to Multi-User Communications

Task 1.1: Traditional FDM, LPF, and the MSE Metric

Question a: Direct Decoding (No Filter)

```
% Task 1.1 Part a)
d1 = 1;
d2 = -1;
f1 = 100;
all_delta_f = [5, 10, 40, 100];
ts = 0.001;

n = 0:1:99;
x1 = d1 * exp (1j * 2 * pi * f1 * n * ts);
for delta_f = all_delta_f
    f2 = f1 + delta_f;
    x2 = d2 * exp (1j * 2 * pi * f2 * n * ts);

    y = x1+x2;
    ybb = y .* exp(-1j * 2 * pi * f1 * n * ts);
    r = ybb;

    mse = mean((real(r) - d1) .^ 2);
    fprintf ('The MSE for delta_f = %d is %f\n', delta_f, mse);
end
```

Figure 1: MATLAB code to Task 1.1 Question a

```
The MSE for delta_f = 5 is 0.500000
The MSE for delta_f = 10 is 0.500000
The MSE for delta_f = 40 is 0.500000
The MSE for delta_f = 100 is 0.500000
```

Figure 2: Result to Task 1.1 Question a

From the result in Figure 2, in direct decoding with no filter, the second terms heavily influence the result so that no matter how far apart of the two signals in modulation, the second term always contributes to a constant MSE of 0.5.

Question b: LPF Receiver

```
% Task 1.1 Part b)
for delta_f = all_delta_f
    f2 = f1 + delta_f;
    x2 = d2 * exp (1j * 2 * pi * f2 * n * ts);

    y = x1+x2;
    ybb = y .* exp(-1j * 2 * pi * f1 * n * ts);

    lpf = fir1(30, 0.1);
    r = filtfilt(lpf, 1, ybb);

    mse = mean((real(r) - d1) .^ 2);
    fprintf ('Part b: The MSE for delta_f = %d is %f\n', delta_f, mse);

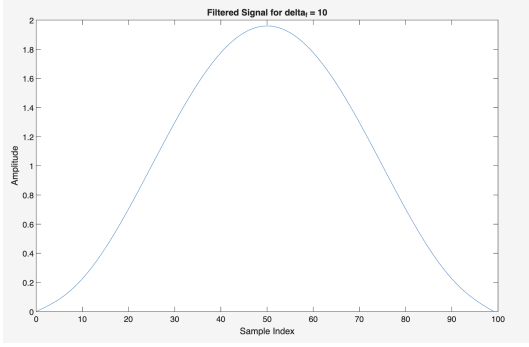
    if delta_f == 10
        figure;
        plot(n, real(r));
        title('Filtered Signal for delta_f = 10');
        xlabel('Sample Index');
        ylabel('Amplitude');
    end
    if delta_f == 100
        figure;
        plot(n, real(r));
        title('Filtered Signal for delta_f = 100');
        xlabel('Sample Index');
        ylabel('Amplitude');
    end
end
end
```

Figure 3: MATLAB code to Task 1.1 Question b (in addition to Figure 1)

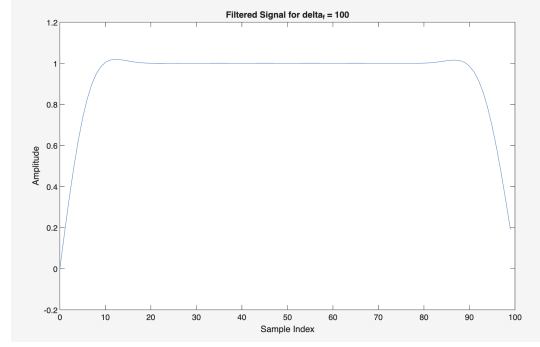
```
Part b: The MSE for delta_f = 5 is 0.490909
Part b: The MSE for delta_f = 10 is 0.464360
Part b: The MSE for delta_f = 40 is 0.138584
Part b: The MSE for delta_f = 100 is 0.044006
```

Figure 4: Result to Task 1.1 Question b

Here, we can see from Figure 4 that as the modulation frequency distance increases, the MSE error decreases. Since our LPF passes frequencies less than 50Hz, when $\Delta f = 5Hz, 10Hz, 40Hz$ interference still exists. However, when $\Delta f = 100Hz$ the LPF completely isolated the effect of $x_2[n]$, giving us a perfect reconstruction of $x_1[n]$ (The slight MSE is due to the non-idealness of LPF used). From Figure 5 we can also clearly see the effect of the LPF on different Δf . Figure 5a shows an interfered $x_1[n]$ due to $x_2[n]$, while figure 5b shows a perfect reconstruction of $x_1[n]$.



(a) $\Delta f = 10 \text{ Hz}$



(b) $\Delta f = 100 \text{ Hz}$

Figure 5: Comparison of the filtered signals $r[n]$ when $\Delta f = 10 \text{ Hz}$ (a) and $\Delta f = 100 \text{ Hz}$ (b).

Question c: Guard Band and spacing

From Question b, we see that a $\Delta f = 100 \text{ Hz}$ has a MSE of roughly 0. This means the Guard Band required for this LPF is around 100Hz. This causes a low spectrum efficiency because each channel needs to occupy 100Hz of the channel, causing too few channels able to communicate at the same time. For $\Delta f = 10 \text{ Hz}$ this LPF completely failed the mission to separate $x_1[n]$ and $x_2[n]$ because $x_2[n]$ lies around 10Hz, which is still in the passband of the low pass filter, making the separation not possible.

Task 1.2: A New Receiver and the Critical Spacing

Question a: The Correlation Receiver

```
% Task 1.2 Part a)
delta_f_sweep = 0:0.5:100;
E = zeros(1, length(delta_f_sweep));

for i = 1:length(delta_f_sweep)
    delta_f = delta_f_sweep(i);

    f2 = f1 + delta_f;
    x2 = d2 * exp (1j * 2 * pi * f2 * n * ts);

    y = x1+x2;
    ybb = y .* exp(-1j * 2 * pi * f1 * n * ts);
    d1_hat = mean(ybb);

    E(i) = abs(d1_hat - d1)^2;
end

figure;
plot(delta_f_sweep, E);
title('Error vs Delta f');
xlabel('Delta f (Hz)');
ylabel('Error');
```

Figure 6: MATLAB code to Task 1.2 Question a (in addition to Figure 1)

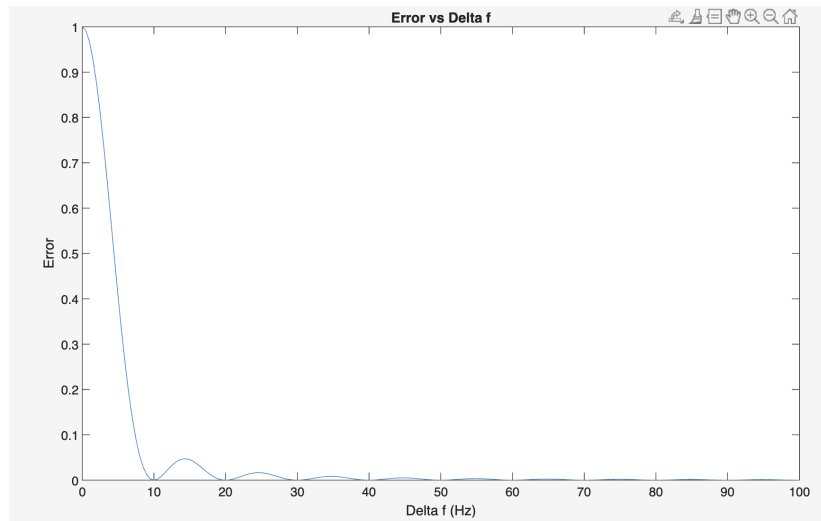


Figure 7: Result to Task 1.2 Question a

Question b: Observation & Discovery

According to Figure 7, we see that the error generally follows a decreasing pattern. More interestingly, the error has a ripple effect, which at every multiple of 10Hz the error is 0, while for non-multiples of 10Hz frequency the error is non-zero (with an overall decreasing trend). Despite $\Delta f < 100Hz$, which through traditional LPF method their frequency spectra overlaps, at certain frequency we can still achieve zero error. The minimum non-zero Δf is $10Hz = \frac{1}{NT_s}Hz$

Question c: The Mathematical Proof of Orthogonality

$$I = \frac{1}{N} \sum_{n=0}^{N-1} e^{j2\pi\Delta f n T_s}$$
$$I = \frac{1}{N} \sum_{n=0}^{N-1} (e^{j2\pi\Delta f T_s})^n$$
$$I = \frac{1}{N} \frac{1 - e^{j2\pi\Delta f T_s N}}{1 - e^{j2\pi\Delta f T_s}}$$

For us to achieve $I = 0$, we need:

$$e^{j2\pi\Delta f T_s N} = 1$$

Use Euler Formula, we get:

$$\begin{cases} \cos(2\pi\Delta f T_s N) = 1 \\ \sin(2\pi\Delta f T_s N) = 0 \end{cases}$$

$$2\pi\Delta f T_s N = 2\pi k, \text{ where } k \in \mathbb{Z}$$

$$\Delta f T_s N = k, \text{ where } k \in \mathbb{Z}$$

$$\Delta f = \frac{k}{NT_s}, \text{ where } k \in \mathbb{Z}$$

Additionally, we need to make sure that the denominator $\neq 0$. Using a similar approach, we deduce that:

$$\Delta f \neq \frac{k}{T_s}, \text{ where } k \in \mathbb{Z}$$

Combining the two equations, we get the final form:

$$\Delta f = \frac{k}{NT_s}, \text{ where } k \in \{1, 2, \dots, N-1\}$$

Task 1.3: Scaling Up to N Users

Question a: Receive signal model

$$y[n] = \sum_{k=0}^{N-1} x_k[n] = \sum_{k=0}^{N-1} d_k e^{j2\pi f_k n T_s} = \sum_{k=0}^{N-1} d_k e^{j2\pi \frac{k}{NT_s} n T_s} = \sum_{k=0}^{N-1} d_k e^{jk \frac{2\pi}{N} n}$$

From the expression, we see that $y[n]$ looks very similar to Inverse Discrete Fourier Transform (IDFT) formula, with only one constant term $\frac{1}{N}$ missing. The IDFT formula is:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{jk \frac{2\pi}{N} n}$$

We see that the receiving step is effectively taking an IDFT (scaled up by a factor of N) to the incoming signals when treating them as a vector.

Question b: Decoding in Receiver

$$\hat{d}_k = \frac{1}{N} \sum_{n=0}^{N-1} y[n] e^{-j2\pi f_k n T_s} = \frac{1}{N} \sum_{n=0}^{N-1} y[n] e^{-j2\pi \frac{k}{NT_s} n T_s} = \frac{1}{N} \sum_{n=0}^{N-1} y[n] e^{-jk \frac{2\pi}{N} n}$$

From the expression, we see that $\hat{d}_k$ looks very similar to the Discrete Fourier Transform (DFT) formula, with only one constant term $\frac{1}{N}$ added. The DFT formula is:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-jk \frac{2\pi}{N} n}$$

We see that the decoding step is effectively taking a DFT (scaled by a factor of $\frac{1}{N}$) to the combined signals $y[n]$.

Question c: The Multi-User Simulation

```
% Task 1.3 Part c)
N = 1024;
input = 2 * randi([0, 1], 1, N) - 1;

modulation = N * ifft(input);

output = 1 / N * fft(modulation);

MSE = mean((output - input).^ 2);

fprintf('\n')
fprintf('MSE for reconstructed signal: %d\n', MSE);
```

Figure 8: MATLAB code to Task 1.3 Question c

```
MSE for reconstructed signal: 1.611663e-32
```

Figure 9: Result to Task 1.3 Question c

According to Figure 9, since the error is very close to 0, it shows that this multi-user communication method works based on DFT and IDFT principles (The small error is due to computer calculation errors and it is essentially zero).

Part 2: Audio Filtering Using FIR and IIR Discrete-Time Filters

Task 2.1: Create and Listen to a Noisy Speech Signal

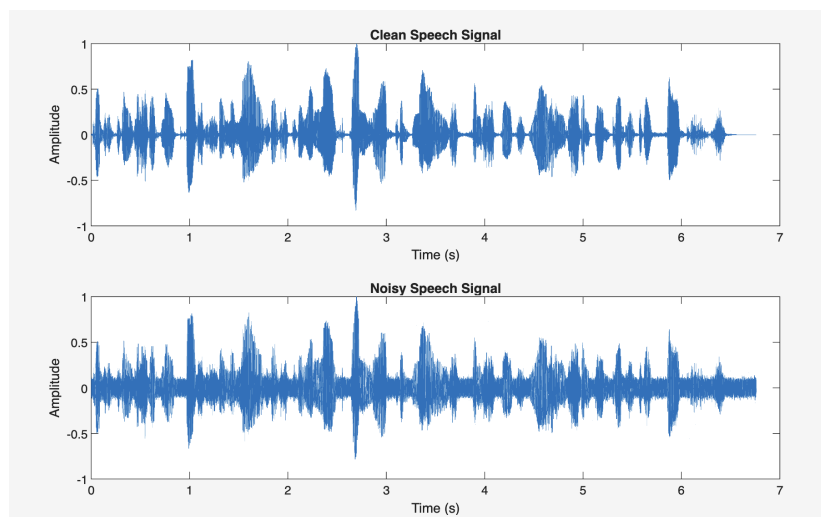


Figure 10: Time-domain plots of the clean and noisy signals

Task 2.2: Frequency-Domain Analysis of the Clean and Noisy Signals

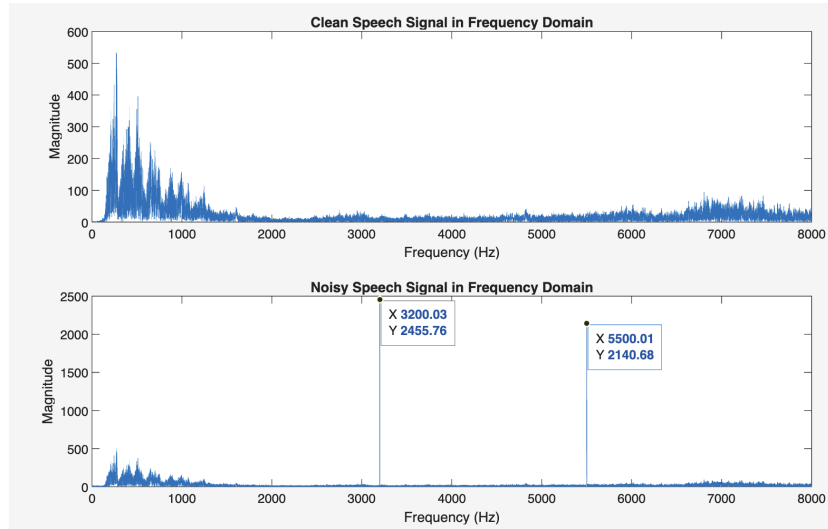


Figure 11: Frequency-domain plots of the clean and noisy signals

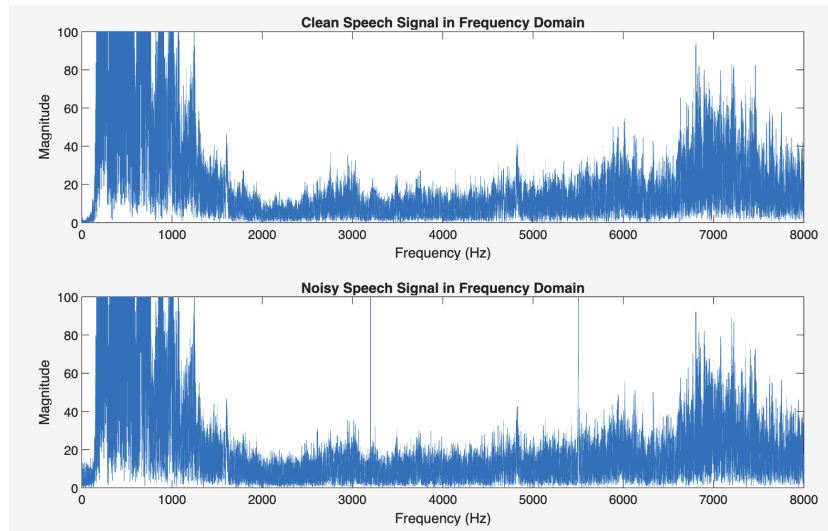


Figure 12: Frequency-domain plots of the clean and noisy signals (Zoomed for better view of broadband white noise contribution)

From Figure 11, we can clearly see the peaks at 3.2kHz and 5.5kHz, which are the contribution done by the 3.2kHz and 5.5kHz sinusoidal noise, respectively. Figure 12 allows us to better see the broadband white noise contribution in the noisy signal, in which they cause the frequency "floor" to be slightly elevated.

Task 2.3: Analysis of the Given FIR and IIR Filters

a) FIR Filter:

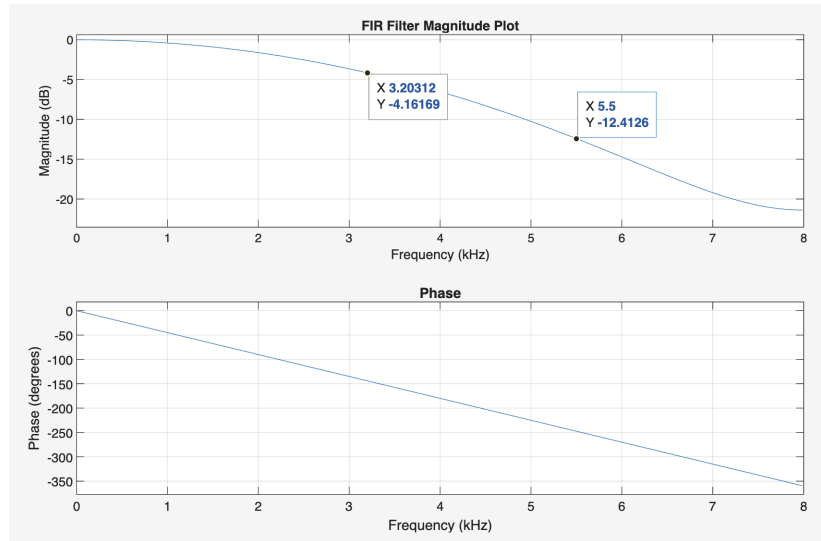


Figure 13: Magnitude response (dB) and phase response (deg) of the FIR filter.

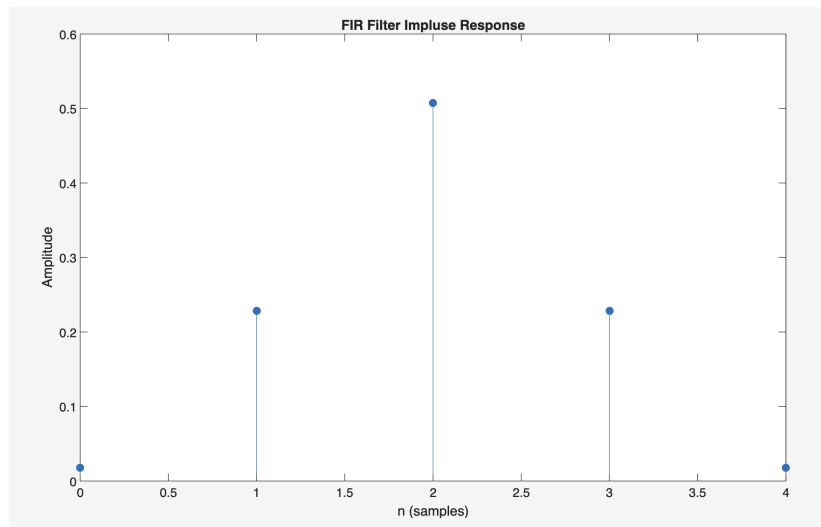


Figure 14: Impulse Response of the FIR filter

From Figure 13, we see the phase response for an FIR Filter is approximately linear. In the magnitude response plot we see that the FIR filter provides an approximately -4.16dB attenuation for frequency around 3.2kHz and an approximately -12.41dB attenuation for frequency around 5.5kHz.

b) IIR Filter:

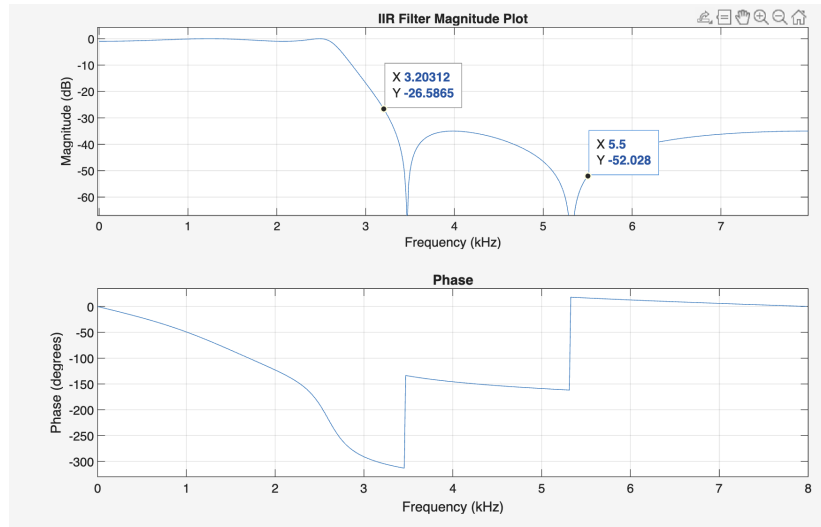


Figure 15: Magnitude response (dB) and phase response (deg) of the IIR filter.

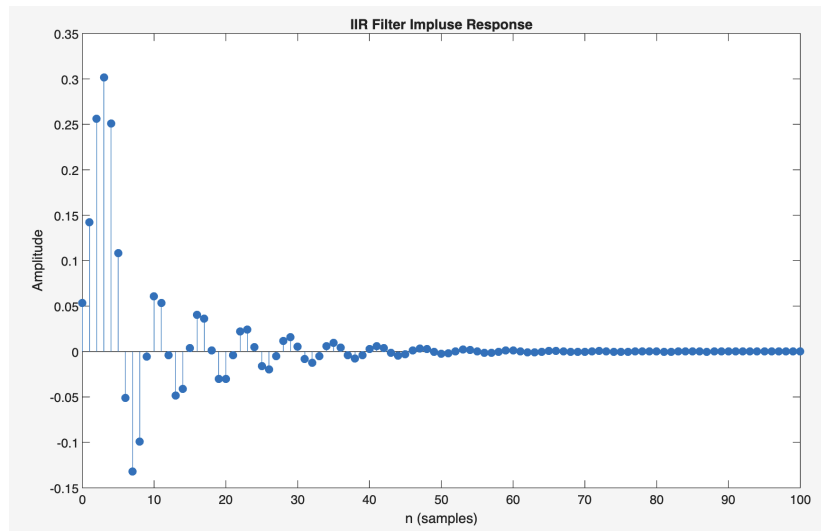


Figure 16: Impulse Response of the IIR filter (up to the first 100 samples)

From Figure 15, we see the phase response for an IIR Filter is nonlinear. In the magnitude response plot we see that the IIR filter provides an approximately -26.59dB attenuation for frequency around 3.2kHz and an approximately -52.03dB attenuation for frequency around 5.5kHz.

Task 2.4: Apply the FIR and IIR Filters to the Noisy Signal

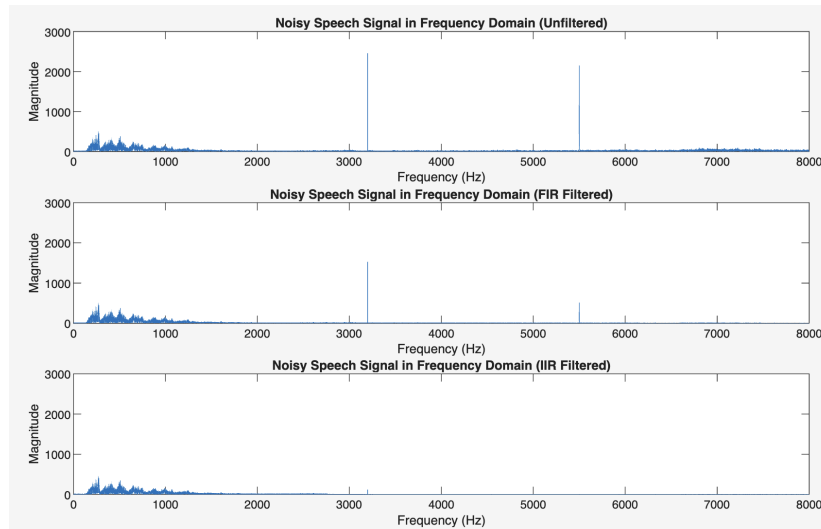


Figure 17: Frequency-domain plots of the filtered and unfiltered noisy signal

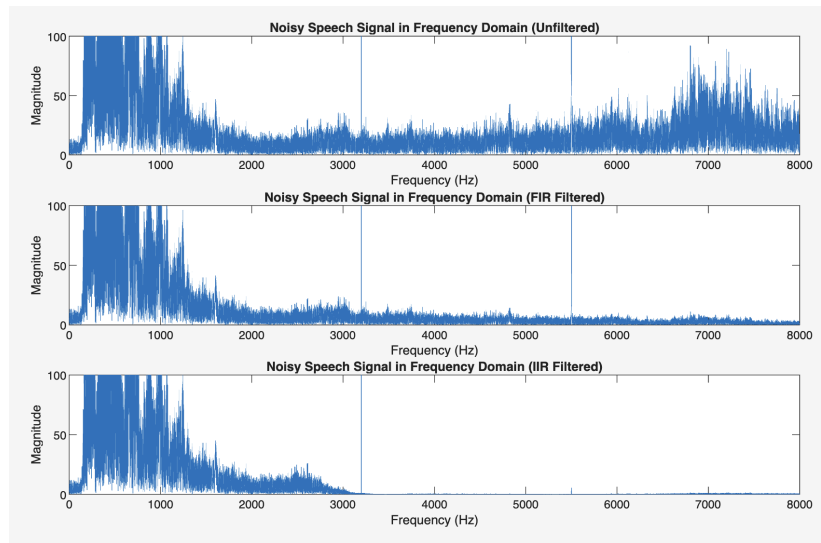


Figure 18: Frequency-domain plots of the filtered and unfiltered noisy signal (Zoomed for better view of broadband white noise contribution)

According to Figure 17 and 18, we see that

1. The IIR filter removes high frequency tones more effectively.
2. The IIR filter has a sharper transition from low frequency to high frequency
3. FIR has a more linear phase than IIR, but IIR are stronger in terms of reducing unwanted noise in higher frequencies. Therefore, the tradeoff is the better linear phase it gets, the weaker the filter is to reduce noise and vice versa.

Task 2.5: Listen to the Filtered Audio

After hearing the filtered audio for both FIR and IIR filters, the observations are:

1. The annoying tones are reduced for both filtered audio, however, the IIR-filtered audio did a better job to remove almost all of the annoying tones from the original noisy audio
2. The broadband noise is reduced for both filtered audio. Both FIR and IIR-filtered audio did a roughly similar job removing some of the broadband noise but still some broadband noise can be heard from the filtered audio.
3. The FIR-filtered audio sounds more natural, more similar to the original voice of the speech. However, the IIR-filtered audio sounds more unnatural, more "masked" and sounds like old radios sounds.
4. The IIR-filtered audio sounds weaker, duller, and slightly distorted compared to the FIR-filtered audio.